

A unified cross-source context enhancement model for multi-source fake news detection

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ABSTRACT

The growing prevalence of misinformation across digital platforms poses a serious threat to public information integrity. While existing fake news detection methods have predominantly focused on single-source data, such approaches often fail to address the complexities introduced by multi-source information originating from diverse platforms. Detecting fake news in this multi-source context remains a relatively underexplored challenge. Traditional cross-domain methods typically transfer domain-invariant features using single-source modeling strategies, yet they neglect the rich contextual cues available across sources—an aspect critical for enhancing detection accuracy. To overcome these limitations, we propose a unified framework that effectively captures global contextual information by jointly leveraging intra-source and inter-source interactions. Our model integrates two principal components: (1) a cross-source global context learning module employing a context-augmented transformer to model long-range dependencies among multi-source instances, and (2) a dual-level contrastive learning mechanism that aligns representations at both local and global levels, reducing inconsistencies across feature spaces and source domains. Extensive experiments conducted on publicly available multi-source datasets demonstrate that our method achieves substantial improvements over existing state-of-the-art approaches. Specifically, it yields an approximate 5% gain in classification accuracy compared to leading models such as LIMFA, highlighting the robustness and effectiveness of our framework in advancing multi-source fake news detection.

1. Introduction

With the emergence of online social media platforms such as Weibo [1], Twitter [2], and a wide range of news portals, the public now has immediate and unprecedented access to information. Yet this convenience comes at a cost: the rapid proliferation of misinformation. Fake news—defined as intentionally fabricated or misleading content—can distort public opinion, undermine trust in institutions, and sow discord among communities. As evidence of this pervasive threat, Weibo alone reported about 82,000 instances of misinformation in 2022 [1].

To date, many approaches to fake news detection [3–8] have focused on analyzing content from a single platform. In contrast, both authentic and fabricated stories often originate from multiple sources, including social networks, instant messaging services, and traditional online news media [9,10], as shown in Fig. 1(a). Relying on a single

source can be insufficient: without considering cross-platform differences in style, presentation, and linguistic cues, it is challenging to distinguish legitimate content from misinformation. This observation raises a key research question: *Can we develop a unified fake news detection model that is both source-agnostic and adaptable across multiple content domains?*

Achieving effective cross-source fake news detection introduces several challenges. First, the inherent heterogeneity of data from various sources complicates the process of feature alignment, information fusion, and modality integration. Although attention-based methods [1,12] have shown promise in handling multimodal information, such approaches often struggle when extended to the complexity of multi-source settings. Second, content manipulation strategies differ across platforms, possibly leading to contradictory or redundant information. These discrepancies introduce noise that can degrade model performance.

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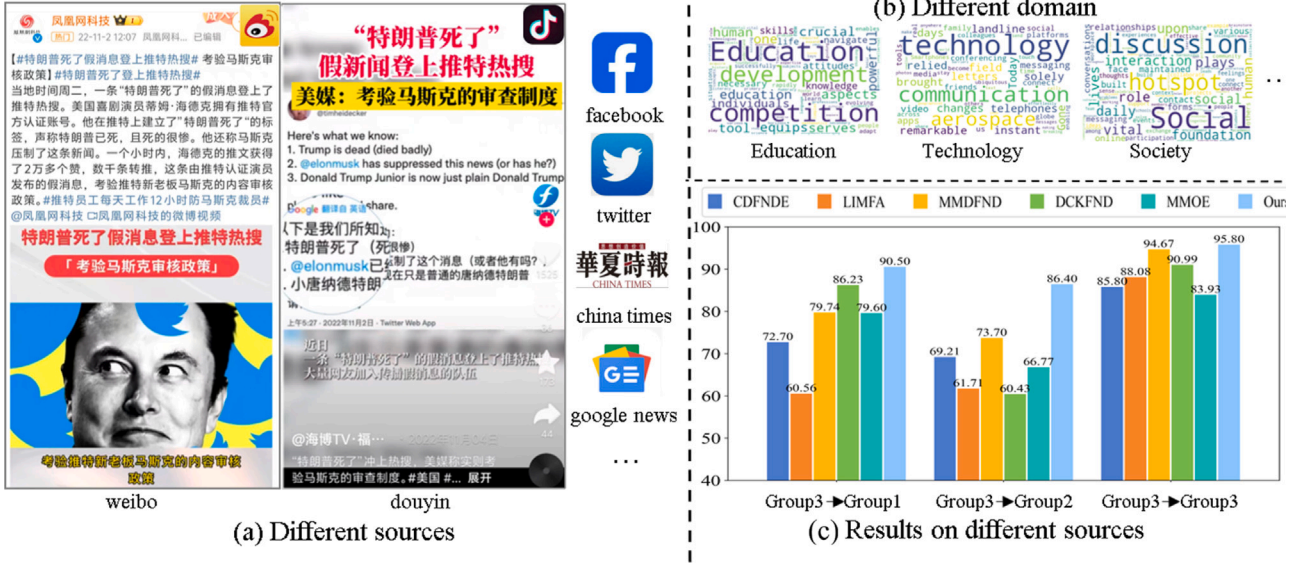


Fig. 1. (a) presents an example of multi-source fake news, originating from three distinct digital platforms. (b) is a word cloud of domains in the multiple-source fake news dataset MCFEND [11]. (c) shows the comparison results of several representative state-of-the-art cross-domain fake news detection models on the MCFEND [11].

Inspired by cross-domain fake news models [13,14], we can naturally question whether these techniques could be directly transferred to cross-source fake news detection tasks. For example, Gong et al. [13] incorporated sample-level and feature-level domain-adaptive representation learning within BERT for sentiment analysis, and Mang et al. [14] leveraged autoencoders for unsupervised adaptation. Though these domain adaptation strategies show promising performance, they struggle when confronted with the more complex inter-source heterogeneity inherent in multi-source fake news detection. In Fig. 1(c), we have conducted the comparison of cross-domain fake news detection models in the setting of a multi-source dataset MCFEND [11]. It is evident that those models, neither the single source nor domain adaptive based, degrade significantly in the multi-source scenario. For example, all methods commonly show superior performance on the intra-group transfer (i.e., Group3 → Group3), but worse on the inter-group transfer. For more experimental analysis, please refer to Section 5.2. Those results confirm that existing cross-domain solutions deteriorate significantly in multi-source scenarios.

To address the challenges of cross-source fake news detection, we introduce a unified cross-source context enhancement framework designed to capture and integrate rich contextual cues across multiple sources. Our core idea is to enhance the learning of global context across various sources so that the model can capture uniform and nuanced cues across sources. To this end, our approach encompasses two core components: cross-source context learning and global context-guided contrastive learning. Specifically, the first component focuses on extracting and refining multimodal features — textual and visual — while incorporating source-specific cues to account for source-specific biases. Meanwhile, we further leverage adversarial training to mitigate the source bias, aiming to extract features that are agnostic to the source, making them more adaptable to detecting fake news across multiple sources.

Secondly, once local features are extracted, the next challenge is integrating information across sources to build a holistic understanding of fake news patterns. To this end, we incorporate a dual-level contrastive learning objective on both local and global features to reinforce robust, consistent, and discriminative representations. Concretely, local contrastive learning occurs within a single data source, enhancing the discrimination of local modal features by semantically aligning text and image modalities within the same source. In contrast, the global context-guided contrastive learning component achieves this

by using a global feature enhancement to enrich each instance with complementary information from other sources.

The key contributions of this work include the following:

- **Unified Cross-Source Detection Framework:** We present a novel framework that enhances contextual understanding across multiple sources, improving the detection of fake news from diverse platforms.
- **Cross-Source Global Context Modeling:** We design a global feature repository and cross-source context augmentation to capture and integrate long-range, inter-source relationships. By incorporating source prototypes, we enable more informed alignment and fusion of features across heterogeneous sources.
- **Dual-Level Joint Contrastive Learning:** We introduce a dual-level contrastive learning strategy that operates on both local and global representations, ensuring robust alignment across modalities and sources. This approach mitigates biases arising from source discrepancies and enhances overall detection performance.

We conduct extensive experiments on publicly available multi-source datasets. Our method substantially outperforms state-of-the-art baselines, with improvements of up to ~5% over the LIMFA model on the Fake News dataset, confirming the efficacy of our proposed approach. Moreover, we curate a new multi-source fake news (MuFake) dataset, and our model still outperforms others with a large margin.

2. Related work

In this section, we provide a comprehensive review of the existing literature on fake news detection, with a focus on single-source approaches and advancements in cross-domain fake news detection.

2.1. Fake news detection

Early efforts in fake news detection primarily focused on single-source analysis [7,15–19]. For example, Qian et al. [20] developed a hierarchical multi-modal contextual attention network to analyze textual data alongside user responses, demonstrating the benefits of integrating semantic information from diverse sources. Similarly, Yang et al. [21] incorporated user engagement metrics, leveraging user profiles and interaction patterns to enhance the identification of fake news.

Recent studies have emphasized the critical role of visual information in fake news detection [3,22]. While traditional unimodal approaches [23,24] achieved notable performance, the integration of multimodal data has been shown to significantly enhance the robustness and accuracy of detection models. For instance, the EANN model [24] combines textual and visual features using a multitask learning framework to support event recognition and fake news detection. Likewise, MKEMN [25] synthesizes textual and visual data with knowledge-aligned embeddings, creating comprehensive multimodal representations of news content.

Despite these advances, multimodal fake news detection still faces challenges. A prominent issue is the cross-modal semantic gap, which refers to the discrepancy between the symbolic pixel-level representations of visual data and the rich semantic information inherent in textual content. This gap complicates the effective alignment of correlated features across modalities. Additionally, existing methods for multimodal data fusion often overlook the heterogeneity of news sources, limiting their scalability in complex, real-world scenarios.

2.2. Cross-domain fake news detection

Cross-domain modeling seeks to enhance generalization by transferring knowledge from one or more domains to a target domain. In fake news detection, cross-domain adaptation is particularly crucial due to the dynamic nature of user behaviors and the evolving patterns of content propagation. Recent work has explored the potential of cross-domain strategies to improve the generalizability and accuracy of fake news detection models.

For example, Mosallanezhad et al. [26] proposed REAL-FND, a cross-domain fake news detection framework that uses reinforcement learning to optimize news representations for better alignment between source and target domains. This model incorporates auxiliary data, such as user feedback and interaction history, to improve adaptability. Similarly, Karami et al. [27] introduced LIMFA, a model designed to handle fake news detection in unseen domains by aligning features across multiple domains. Shu et al. [28] utilized a prompt-based strategy to tailor models for specific domain requirements, further enhancing the efficacy of cross-domain fake news detection.

While these approaches highlight the promise of cross-domain adaptation, most focus on feature alignment within a single information domain. Challenges such as the heterogeneity of online platforms, regional differences, and temporal shifts in data distribution remain underexplored. In particular, the problem of identifying fake news across multiple sources has received limited attention.

3. Problem definition

Fake news detection is typically defined as a binary classification task, where the objective is to assess the credibility of information. Formally, given input data $D = \{(x_i, y_i)\}$ where $x_i \in \mathcal{X}$ represents the input multimodal news (containing images and texts), and $y_i \in \{0, 1\}$ denotes the label indicating authenticity (with 1 for true news and 0 for fake news), the aim is to train a model Φ that can accurately determine the authenticity of the news items. In the traditional setting [29–31], it is assumed that the input data is drawn from a single source, i.e., $\mathcal{X} \in S$, where S represents the set of possible sources, and $|S| = 1$. However, in the context of multi-source fake news detection, we relax this assumption, allowing $|S| > 1$, meaning the source of $x_i \in S$ is variable, originating from multiple sources such as Weibo [32], China Daily [33], and Gossipcop [34]. This setting demands that the model capture patterns of fake news across diverse sources.

4. Methods

In this section, we present our proposed method for cross-source multimodal fusion aimed at enhancing fake news detection. The model processes both textual and visual data from multiple sources, employing four key components to achieve robust misinformation classification, as shown in Fig. 2. First, the multi-source feature encoding component utilizes dedicated encoders to transform text and image inputs into feature representations, aligning modalities through Dynamic Time Warping (DTW) to ensure multimodal coherence. Next, the cross-source global context learning module incorporates source-specific cue injection to capture both intra-source and inter-source contextual relationships, mitigating source biases in misinformation detection. To further refine feature representations, a dual-level contrastive learning module is introduced, which enhances both local and global feature consistency while distinguishing cross-source characteristics. Finally, the cross-source multimodal decoder integrates these refined features and classifies them into predictive probabilities, thereby determining the likelihood of misinformation. By systematically combining these components, our approach effectively harnesses multimodal and multi-source information, addressing the inherent challenges of cross-domain fake news detection and improving classification accuracy.

4.1. Multi-source feature encoding

The task of cross-source fake news detection takes as input heterogeneous data sources, and each source provides multimodal information, including both text and image modalities. Formally, given the diverse multimodal information with text and image modalities from heterogeneous sources, we formalize the feature extraction process for both modalities as follows:

$$h_a^\mu = \mathcal{E}_\mu(x_a^\mu) \quad (1)$$

where $h_a^\mu \in \mathbb{R}^{\mu \times d}$ represents the embedding feature of the modality μ extracted from data source $a \in S$ indicating the data source, where $\mathcal{E}_\mu$ is the encoder for modality μ , and x_a^μ is the raw input of modality μ from data source a . Here, $\mu \in \{t, v\}$, with t denoting the text modality (using a pre-trained BERT [6]) and v representing the image modality (using a ResNet [35]).

To address the issue of inconsistent sequence lengths across different modalities, we use Dynamic Time Warping (DTW) [36] to align the textual features $h_{a,i}^t$ and visual features $h_{a,i}^v$. Specifically, DTW constructs a correlation matrix between the text and image features using a dynamic programming algorithm, where the core computation involves the calculation of the cumulative distance matrix:

$$D(p, q) = \|h_{a,i}^t(p) - h_{a,i}^v(q)\|^2 + \min\{D(p-1, q), D(p, q-1), D(p-1, q-1)\} \quad (2)$$

Here, $h_{a,i}^t(p)$ represents the feature vector of the text modality at sequence position p for the i th sample in data source a , and $h_{a,i}^v(q)$ is the corresponding feature vector of the image modality at position q . We first compute the squared Euclidean distance between these two features, then update the distance recursively by taking the minimum cumulative distance from the three neighboring directions: above $D(p-1, q)$, left $D(p, q-1)$, and diagonal $D(p-1, q-1)$. Besides, we also examine other alternative alignment strategies, e.g., Connectionist Temporal Classification CTC [37]. The detailed discussion is presented in the experiment section.

To balance computational efficiency and alignment accuracy, we introduce the Sakoe-Chiba Band constraint during the recursion, limiting the path search within a band of width r , meaning that the distance accumulation is only performed when $|p - q| \leq \ell$. After obtaining the entire matrix, the optimal alignment path $\mathcal{P}$ is determined by backtracking from the endpoint $D(M, N)$ along the minimum cumulative path. Finally, interpolation based on this path is applied to the original

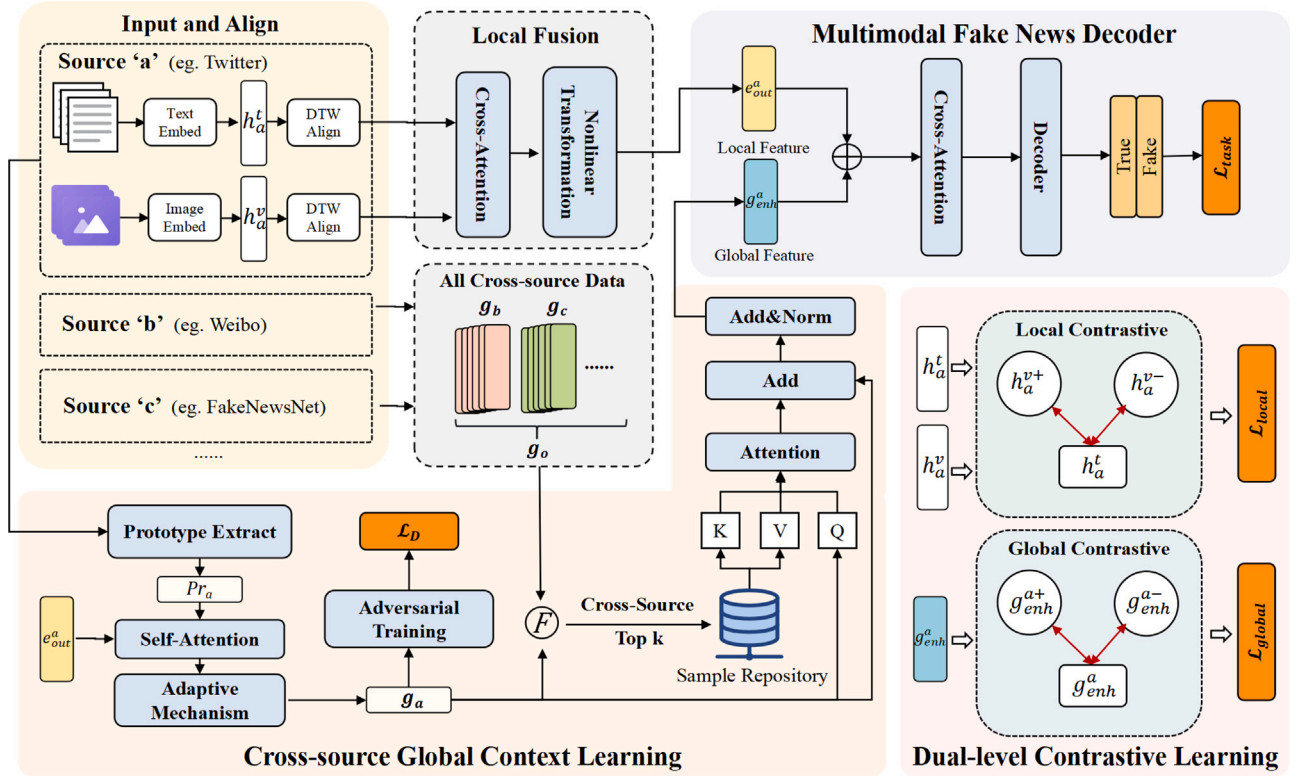


Fig. 2. The overview of our model, which consists of four key components. First, a feature encoding module embeds and aligns multimodal data using Dynamic Time Warping (DTW). Next, the model employs global context learning, which acts like a virtual library borrowing relevant information from other sources to strengthen the model's understanding of each news item. This process helps extract relationships within and between data sources, thereby reducing source bias. A dual-level contrastive learning module then captures both local and global feature consistencies to refine representations. Finally, a multimodal decoder combines these features to classify fake news by generating predictive probabilities.

feature sequences, resulting in length-aligned text features $\hat{h}_{a,i}^t$ and image features $\hat{h}_{a,i}^v$. Notably, compared to other alignment strategies like Connectionist Temporal Classification (CTC), the advantage of DTW lies in its ability to handle non-monotonic alignments, making it better suited for the complex temporal relationships between multimodal features.

Regarding implementation details, we restrict the path search range using the Sakoe-Chiba Band constraint [38], setting the bandwidth to 26 (calculated as $r = \lceil 0.2 \times \max(M, N) \rceil$), and strictly enforce the condition $\ell = 26$ during dynamic programming recursion. The cumulative distance matrix is normalized by path length ($D(p, q) \leftarrow D(p, q)/(p + q)$), and feature alignment is achieved via linear interpolation based on the optimal path $\mathcal{P}$. More details are provided in the section of implementation details.

Subsequently, we fuse the features of the two modalities through a cross-attention mechanism, calculated as:

$$e^{a,i} = \text{softmax} \left(\frac{(W_q \hat{h}_{a,i}^t)(W_k \hat{h}_{a,i}^v)^T}{\sqrt{d}} \right) W_v \hat{h}_{a,i}^v \quad (3)$$

where both the text feature $\hat{h}_{a,i}^t$ and the image feature $\hat{h}_{a,i}^v$ are dimensioned as $\mathbb{R}^{L \times d}$, L represents the sequence length of the text and image modalities, and d is the embedding dimension. W_q , W_k , and W_v are three projection layers for the query, key, and value. Finally, we use an MLP on $e^{a,i}$ by:

$$e_{out}^{a,i} = \sigma(W \cdot e^{a,i} + b) \quad (4)$$

where W is the weight matrix for the linear transformation, b is the bias term, and σ is the nonlinear activation function. In this work, we denote $e_{out}^{a,i}$ as the local features because they only capture the representations in a single source.

4.2. Cross-source global context learning

Intuitively, the local features extracted from multimodal encoders primarily capture modality-specific information but lack source-specific characteristics. However, different sources exhibit distinct data distributions, as highlighted in [11], despite sharing the same modalities. Consequently, it is essential to incorporate source identity information into feature representations to enable the detection model to learn and differentiate fake news patterns across various sources. To achieve this, we first extract a prototype for each source and employ a cross-attention mechanism to embed this prototype into the local feature representation of each sample, thereby enriching the model's ability to capture source-specific variations and improving cross-source fake news detection.

4.2.1. Source-specific cues injection

Initially, for each data source a , we compute the source prototype Pr_a by taking the learnable weighted average of the local features of all samples as:

$$Pr_a = \epsilon \cdot \frac{1}{N_t} \sum_{i=1}^{N_t} \hat{h}_{a,i}^t + (1 - \epsilon) \cdot \frac{1}{N_v} \sum_{j=1}^{N_v} \hat{h}_{a,j}^v \quad (5)$$

where $\hat{h}_{a,i}^t$ and $\hat{h}_{a,j}^v$ represent the local embedding features of the i th text sample and the j th image sample from data source a , respectively. N_t and N_v are the total number of text and image modality samples in data source s_a , and ϵ is a learnable weight parameter to balance text and image modal features.

Then, for each sample, we inject the source-identical cue, i.e., the prototype Pr_a , to specify each sample's source. To this end, we employ a self-attention mechanism to compute the relevance weights $\alpha_{a,i}$ of

each sample concerning the source prototype. This process can be written as:

$$\alpha_{a,i} = \text{softmax} \left(\frac{(W_q e_{out}^{a,i})(W_k P r_a)^T}{\sqrt{d}} \right) \quad (6)$$

Subsequently, we leverage an adaptive mechanism that dynamically tailors the representation of source prototypes for individual samples, defined as follows:

$$g_{a,i} = \text{sigmoid}(\omega \cdot \alpha_{a,i}) \cdot P r_a + (1 - \text{sigmoid}(\omega \cdot \alpha_{a,i})) \cdot (e_{out}^{a,i}) \quad (7)$$

where $g_{a,i}$ denotes the feature with the injected source-specific cues, while ω serves as a tuning parameter, which is smoothed by a sigmoid function.

To force $g_{a,i}$ to align with its corresponding source distribution, we further impose an adversarial training, widely used in the domain adaptation [13,21]. Specifically, we train a source discriminator $D(\cdot)$ to judge which source the feature $g_{a,i}$ derives from. When the discriminator cannot judge its source, the features from different sources are well-aligned. Formally, the discrimination loss is written as:

$$\mathcal{L}_D = \mathbb{E}_{e \sim p_{data}(x)} [CE(D(e^i), o^i)] + \mathbb{E}_{g \sim p_g(g)} [CE(1 - D(g^i), o^i)] \quad (8)$$

where $o^i \in S$ denote the source identity of the i th sample and $CE(\cdot)$ is the cross-entropy function.

4.2.2. Cross-source context augmentation

The main advantage of cross-source fake news detection is that it integrates cross-source information to determine the genuineness of news. Our core idea involves augmenting single-source knowledge by integrating relevant data from diverse sources. Thus, we propose a cross-source context augmentation transformer, which enhances the contextual understanding of a given sample by embedding information from similar news originating from various sources. This augmentation can be divided into cross-sample repository construction and cross-source context embedding.

Cross-Source Sample Repository Construction. This step aims to find the most relevant news across different sources, given query news. Specifically, we identify the most relevant samples from other data sources. This relevance is measured through the computation of cosine similarity F between the global feature $g_{a,i}$ of the sample and those from other sources:

$$F(g_{a,i}, g_{o,j}) = \frac{g_{a,i} \cdot g_{o,j}}{\|g_{a,i}\| \|g_{o,j}\|}, \quad o \in \{S \setminus a\} \quad (9)$$

Here, $F(g_{a,i}, g_{o,j})$ represents the similarity between two global features. In practice, we select the k most relevant samples from the two data sources. Hence, we can write the constructed cross-source repository $\Omega(g_{a,i})$ as:

$$\Omega_i^a = \{v_j \mid v_j \in \text{Top}_k(S(g_{a,i}, :))\} \quad (10)$$

where Top_k is a function to return the indexes of the top k most relevant samples according to the similarity. For the other source, we use the same way to obtain Ω_i^o where $o \in S \setminus a$.

Cross-source Context Embedding. With the top- k relevant cross-source samples, we directly treat them as the long-context to augment the global feature. The motivation behind this is that we believe that the context information across different sources can provide crucial clues to alleviate the ambiguity or source bias in cross-source fake news detection.

Given the set of similar samples: $\overline{\Omega_i^a} = \{g_o^{v_1}, g_o^{v_2}, \dots, g_o^{v_k}\}$, the sample $g_{a,i}$ interacts with every sample in the cross-source sample set $\overline{\Omega_i^a}$ via a cross-attention mechanism via:

$$f_{a,i} = \text{cross_attention} \left(g_{a,i}, \overline{\Omega_i^a}, \overline{\Omega_i^a} \right) \quad (11)$$

where $g_{a,i}$ is the query and the top k samples $\overline{\Omega_i^a}$ are the key and value in the cross attention. Subsequently, we fuse the long context with the global feature by:

$$g_{enh}^{a,i} = \phi(f_{a,i} + g_{a,i}) \quad (12)$$

where ϕ is a normalization layer.

4.3. Dual-level local and global contrastive learning

Contrastive learning enhances the representations by effectively reducing the distance between positive samples and the anchor, while simultaneously increasing the distance to negative samples. Traditional contrastive learning [39] often focuses on local, small-batch sample comparisons and does not fully utilize global information. In this work, we not only engage in local modal contrastive learning within single data sources but also introduce global context-guided contrastive learning to enhance the model's discriminative ability. This is to ensure that similar news from the same source (local level) and different sources (global level) are grouped together in the feature space, while dissimilar news are pulled apart, thus enhancing the model's discriminative ability.

4.3.1. Local contrastive learning

This module occurs within a single data source, enhancing the discrimination of local modal features by semantically aligning text and image modalities within the same source. Specifically, given the text modality feature $\hat{x}_a^t$ as an anchor, the positive sample set P_l includes image modality features $\hat{x}_a^{v+}$ from the same label, while the negative sample set N_l includes image modality features $\hat{x}_a^{v-}$ from different labels. The local contrastive loss is expressed using the InfoNCE [40] loss as follows:

$$\mathcal{L}_{\text{local}} = -\log \frac{\exp(\hat{h}_a^t \cdot \hat{h}_a^{v+} / \tau)}{\exp(\hat{h}_a^t \cdot \hat{h}_a^{v+} / \tau) + \sum_{\hat{h}_a^{v-} \in N_l} \exp(\hat{h}_a^t \cdot \hat{h}_a^{v-} / \tau)} \quad (13)$$

where τ is a temperature parameter that controls the sensitivity of the contrastive learning process.

4.3.2. Global contrastive learning

Global contrastive learning is conducted across data sources, enabling the model to capture cross-source semantic consistency and nuanced differences between news from different sources, enhancing the cross-source discriminative ability for fake news detection. Unlike local contrastive learning, global contrastive learning uses the global features of a single sample as the anchor, constructing positive and negative sample pairs across data sources. Specifically, given the global feature $g_{enh}^{a,i}$ of a single sample as an anchor, the positive sample set P_g includes global features $g_{enh}^{a,i+}$ from the same news label across data sources, while the negative sample set N_g includes global features $g_{enh}^{a,i-}$ from different news labels. The global contrastive loss formula is as follows:

$$\mathcal{L}_{\text{global}} = -\log \frac{\exp(g_{enh}^{a,i} \cdot g_{enh}^{a,i+} / \tau)}{\exp(g_{enh}^{a,i} \cdot g_{enh}^{a,i+} / \tau) + \sum_{g_{enh}^{a,i-} \in N_g} \exp(g_{enh}^{a,i} \cdot g_{enh}^{a,i-} / \tau)} \quad (14)$$

4.4. Cross-source multimodal decoder

In this section, we utilize a cross-source multimodal decoder for fake news detection, which incorporates multimodal fusion from both local and global perspectives to capture comprehensive cross-source multimodal feature representations. Initially, we integrate the local and global features through a concatenation operation.

$$h_{\text{fusion}}^i = \phi_n([e_{out}^{a,i}, g_{enh}^{a,i}]) \quad (15)$$

The final classification of fake news is accomplished through the transformer decoder MMDecoder with a binary classification head,

Table 1

Performance comparison with different methods on the multi-source dataset **MCFEND**. We divide the methods into two categories: uni-source, cross-domain. All benchmark models were trained with the news from all sources and evaluated on the test set of specified groups. The highest and second-highest results in each source are highlighted in bold and underlined, respectively.

Category	Method	Group 1		Group 2		Group 3		Overall	
		Acc	F1	Acc	F1	Acc	F1	Acc	F1
Uni-source	TextCNN [5]	0.798	0.523	0.725	0.509	0.666	0.694	0.727	0.695
	VGG-19 [4]	0.701	0.464	0.624	0.455	0.577	0.546	0.630	0.544
	VS [3]	0.666	0.415	0.601	0.579	0.620	0.675	0.625	0.647
	RA [1]	0.846	0.509	0.653	0.549	0.817	0.829	0.783	0.747
	CAR [7]	0.830	0.585	0.751	0.569	0.696	0.755	0.749	0.763
	EANN [41]	0.873	0.567	0.670	0.602	0.840	0.890	0.798	0.809
	BERT [6]	0.934	0.535	0.858	0.614	0.803	0.801	0.862	0.800
	RoBERTa [8]	<u>0.939</u>	0.518	<u>0.860</u>	0.501	0.844	0.844	0.880	0.827
	MVAE [42]	0.898	0.532	0.706	0.582	0.870	0.850	0.834	0.783
	SpotFake [43]	0.897	<u>0.626</u>	0.706	0.674	0.873	0.847	0.827	0.771
	MKEMN [25]	0.889	0.573	0.696	0.616	0.854	0.800	0.824	0.821
	SAFE [44]	0.897	0.581	0.697	0.616	0.857	0.796	0.827	0.833
	CLIP [12]	0.891	0.534	0.691	0.573	0.860	0.852	0.825	0.781
	MVNN [45]	0.907	0.588	0.701	0.626	0.860	0.821	0.832	0.844
	CAFE [46]	0.926	0.563	0.857	0.675	0.875	0.875	0.887	0.845
	REAL-FND [26]	0.925	0.601	0.728	0.675	<u>0.937</u>	<u>0.937</u>	<u>0.922</u>	<u>0.916</u>
Cross-domain	MMOE [47]	0.876	0.675	0.794	0.660	0.825	0.772	0.838	0.789
	DCKFND [48]	0.873	0.657	0.828	0.688	0.897	0.881	0.864	0.847
	CDFND [28]	0.921	0.684	0.711	0.645	0.759	0.712	0.879	0.837
	LIMFA [27]	0.930	0.596	0.843	0.661	0.864	0.791	0.887	0.856
	MMDFND [49]	0.845	0.676	0.834	<u>0.695</u>	0.881	0.887	0.859	0.858
	GAMED [50]	0.931	0.589	0.855	0.610	0.915	0.887	0.912	0.877
	TSHCL [51]	0.855	0.664	0.783	0.542	0.830	0.851	0.831	0.845
Ours		0.951	0.720	0.895	0.705	0.958	0.946	0.957	0.943

which transforms the fused features h_{fusion} into prediction probabilities q_i :

$$q_i = \text{MMDecoder}(h_{\text{fusion}}^i) \quad (16)$$

The classification loss for fake news classification $\mathcal{L}_{\text{task}}$ is based on binary cross-entropy loss as

$$\mathcal{L}_{\text{task}} = - \sum_{i=1}^N y_i \cdot \log(q_i) \quad (17)$$

where y_i is the ground-truth label. Finally, the overall training objective combines the cross-entropy loss with contrastive learning loss, as well as domain adaptation adversarial loss, with the total loss formula being:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \alpha \mathcal{L}_{\text{local}} + \beta \mathcal{L}_{\text{global}} + \gamma \mathcal{L}_{\text{D}} \quad (18)$$

where α , β and γ adjust the weights of each loss respectively.

5. Experiments

5.1. Datasets and settings

MCFEND [11] is the first multi-source benchmark dataset specifically designed for the detection of Chinese fake news. It aggregates 23,789 news sourced from a variety of origins, including social platforms, messaging applications, and traditional online news media. Following [11], we divided the dataset into three groups representing different source types. Group 1 comprises 7959 news from 9 Chinese fact-checking organizations. Group 2 includes 6702 articles verified by 4 English-speaking fact-checking entities, such as Politifact and FakeNewsCorpus. Group 3 encompasses 9128 news from the Weibo-21 dataset, collected from the Weibo platform.

Multi-source Fake News (MuFake) is a newly constructed dataset comprising three distinct source types, totaling approximately 46,430 news articles. Group 1: Twitter dataset, which includes 13,253 articles collated by the MediaEval Verifying Multimedia Use task [2], specifically curated to exclude video-based tweets. Group 2: Weibo dataset, which comprises 9528 articles gathered by [1], featuring verified news

from prestigious sources such as Xinhua News Agency alongside falsified reports identified via the official Weibo debunking system. Group 3: FakeNewsNet dataset [34], proposed by Shu et al. contains about 20,900 news articles from fact-checking websites GossipCop [34] and PolitiFact [34]. News in the FakeNewsNet is verified by professional journalists and experts.

TALLIP[29] is a multi-domain fake news dataset with 4900 news articles covering various fields such as business, education, technology, entertainment, politics, sports, and celebrity.

Baselines. To evaluate our model, we group baseline models for fake news detection into two categories: uni-source and cross-domain approaches. The uni-source models include CAR [7], VGG-19 [4], TextCNN [5], VS [3], BERT [6], RoBERTa [8], RA [1], CLIP [12], EANN [41], MVAE [42], SAFE [44], SpotFake [43], CAFE [46], REAL-FND [26], MKEMN [25], and MVNN [45]. The cross-domain multimodal models include CDFND [28], LIMFA [27], MMDFND [49], DCKFND [48], MMOE [47], GAMED [50] and TSHCL [51]. Our proposed model, denoted as CFE-CL, is also evaluated.

Implementation Details. For feature extraction, we encode the input text into 256-dimensional embeddings using a pretrained BERT model [6], and standardize image dimensions to 224×224 pixels through center-cropping, followed by employing the ResNet-34 model [35] to extract features. We use Dynamic Time Warping [52] for feature alignment, combined with a cross-attention mechanism and a three-layer fully connected network for integration, each layer consisting of 64 hidden units. The global context is learned via a cross-source adversarial training strategy, aligning features across sources with a discriminator comprised of three layers and a learning rate of $1e^{-4}$. The top-k number of nearest neighbors for enhancing global context learning is set at 7. The temperature value is fixed at 0.5. The parameters for the total loss function are set as $\alpha = 0.2$, $\beta = 0.2$ and $\gamma = 0.4$. During training, we use an Adam optimizer with a learning rate of 2^{-5} , a batch size of 32, and a dropout rate of 0.5. All experiments are executed on Intel(R) Xeon(R) Platinum 8462Y+ CPU, 2 NVIDIA A800-SXM4-80GB GPUs, and 1TB RAM. Model performance is evaluated by accuracy (ACC) and F1 score (F1).

Table 2

Performance comparison of different benchmarks on the multi-source dataset **MuFake**. All benchmark models were trained with the news from all sources and evaluated on the test data of specified groups. The highest and second-highest results in each source are highlighted in bold and underlined, respectively.

Category	Methods	Group 1		Group 2		Group 3		Overall	
		Acc	F1	Acc	F1	Acc	F1	Acc	F1
Uni-source	TextCNN [5]	0.594	0.559	0.712	0.586	0.665	0.571	0.591	0.650
	VGG-19 [4]	0.625	0.524	0.677	0.552	0.653	0.516	0.670	0.614
	VS [3]	0.591	0.499	0.716	0.530	0.645	0.612	0.647	0.595
	RA [1]	0.648	0.566	0.743	0.543	0.709	0.676	0.675	0.687
	CAR [7]	0.774	0.561	0.776	0.549	0.733	0.661	0.733	0.727
	EANN [41]	0.751	0.600	0.855	0.569	0.817	0.713	0.777	0.716
	BERT [6]	0.719	0.551	0.846	0.584	0.769	0.668	0.772	0.753
	RoBERTa [8]	0.758	0.586	0.856	0.566	0.777	0.654	0.793	0.774
	MVAE [42]	0.816	0.524	0.858	0.591	0.838	<u>0.826</u>	0.787	0.740
	SpotFake [43]	0.808	0.656	0.900	0.624	0.865	0.778	0.839	0.810
	MKEMN [25]	0.763	0.602	0.863	0.564	0.791	0.712	0.792	0.786
	SAFE [44]	0.798	0.690	0.899	0.560	0.866	0.606	0.826	0.805
	CLIP [12]	0.701	0.602	0.809	0.563	0.776	0.608	0.737	0.715
	MVNN [45]	<u>0.837</u>	0.670	0.880	0.545	0.866	0.787	0.870	0.785
	CAFE [46]	0.820	0.609	0.876	0.591	0.807	0.809	0.838	0.807
	REAL-FND [26]	0.801	0.701	0.877	0.598	0.859	0.743	0.818	0.794
Cross-domain	MMOE [47]	0.825	<u>0.711</u>	0.887	<u>0.655</u>	<u>0.880</u>	0.815	0.876	0.852
	DCKFND [48]	0.812	0.688	0.874	0.642	0.871	0.811	0.859	0.849
	CDFND [28]	0.784	0.667	0.868	0.601	0.865	0.731	0.814	0.758
	LIMFA [27]	0.815	0.698	0.891	0.653	0.868	0.807	0.839	0.835
	MMDFND [49]	0.836	0.705	0.886	0.649	0.867	0.794	0.868	0.751
	GAMED [50]	0.835	0.689	0.894	0.611	0.870	0.788	0.836	0.801
	TSHCL [51]	0.832	0.694	0.891	0.642	0.870	0.807	0.832	0.831
Ours		0.838	0.783	0.914	0.750	0.899	0.873	0.889	0.884

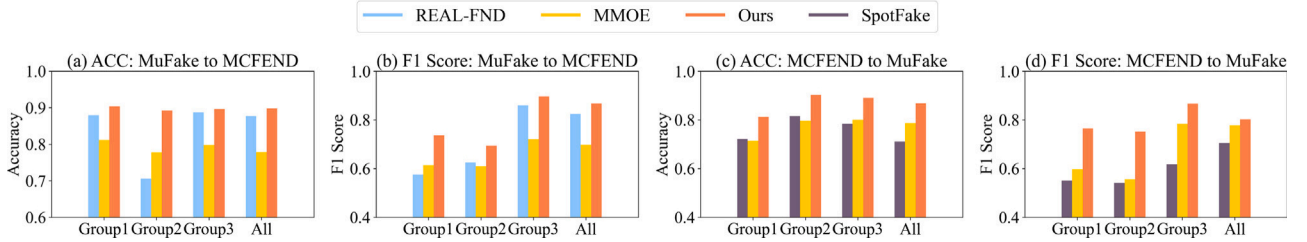


Fig. 3. Performance of our model compared to two other models in different combinations of training and testing datasets. Comparative Performance Analysis Across Training and Testing Dataset Combinations. (a). Accuracy of models trained on MuFake and evaluated on MCFEND. (b). F1 Score of models trained on MuFake and evaluated on MCFEND. (c). Accuracy of models trained on MCFEND and evaluated on MuFake. (d). F1 Score of models trained on MCFEND and evaluated on MuFake.

5.2. Experimental results

5.2.1. Main results

Tables 1, 2 presents the performance comparison of our model against baseline models on the multi-source datasets **MCFEND** and **MuFake**, respectively. All models were trained using news data from all source groups and evaluated on the test sets of specified groups. From the tables, we could obtain the following observations:

(1) **Performance Improvements:** Our model demonstrated superior performance in tests, including singular and multiple data sources, particularly when compared with other popular multimodal learning models such as LIMFA [27] and SpotFake [43]. For example, in a Group 2 assessment, the average Macro F1 score of GFE-CL was about 9.7% higher than that of LIMFA [27], a significant performance improvement that not only proves the practicality of GFE-CL in fake news detection tasks but also its capability in handling complex data.

(2) **Stability Analysis:** Despite the inherent heterogeneity and volatility in a multi-source data environment, the GFE-CL, not on the table model is able to maintain high overall performance while demonstrating relatively higher stability. Particularly in the MuFake dataset, its Macro F1 score variation is significantly lower than that of other comparable high-performance models, with a reduction of 7.3% compared to the DCKFND model. This underscores GFE-CL's exceptional adaptability and reliability in multi-source context.

(3) **Capability in Handling Complex Data:** Within the most challenging data groups identified — Group 2 of the MCFEND dataset [11] and Group 1 of the MuFake dataset — GFE-CL demonstrated pronounced performance superiority. For instance, in MCFEND's Group 2, GFE-CL sustained a Macro F1 score of 0.705 amidst challenging conditions, reflecting its exceptional robustness and adaptability. Similarly, in Group 1 of the MuFake dataset [1,2,11], GFE-CL achieved a Macro F1 score of 0.783, which substantially exceeds the 0.675 scores observed for the baseline model CAFE [46]. This indicates a robust capacity to process and analyze high-complexity data structures effectively.

5.2.2. Robustness results

To evaluate the GFE-CL model's performance in handling previously unencountered news sources, we conducted a series of evaluations on unseen source fake news detection. The experimental setup involved training the GFE-CL model on the MCFEND dataset [11] and testing it across various groups within the MuFake dataset [1,2,11], and vice versa. To ensure a comprehensive evaluation, we also selected the well-performing unimodal and multimodal models from earlier assessments for comparison to validate the performance of GFE-CL in new environments. As shown in Fig. 3, among the models tested and trained on different datasets, the GFE-CL model outperforms the other two models in terms of accuracy and F1 score, with the compared models being the two least performing models among all. The difference in the heights of the bars in the chart highlights the advantages of our GFE-CL.

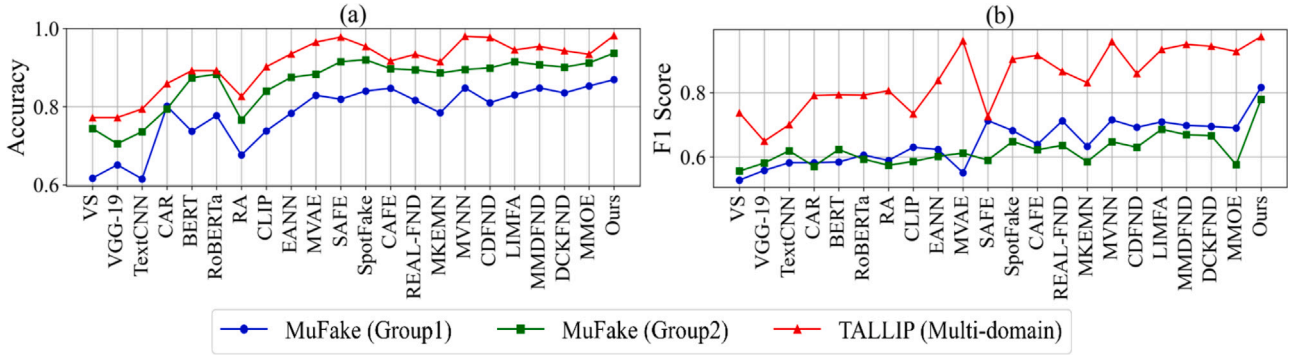


Fig. 4. The results on the single source setting. (a). Accuracy comparison of various models across three single-source datasets. (b). F1 Score comparison of different models across three single-source datasets.

Table 3

Ablation study on the effectiveness of each component.

Methods	Test	MCFEND		MuFake	
		Acc	F1	Acc	F1
w/o DTW [52]	Group 1	0.937	0.702	0.823	0.762
	Group 2	0.884	0.683	0.902	0.728
	Group 3	0.939	0.934	0.886	0.854
	Overall	0.943	0.930	0.876	0.792
with CTC [37]	Group 1	0.914	0.695	0.800	0.745
	Group 2	0.885	0.694	0.880	0.744
	Group 3	0.945	0.939	0.876	0.843
	Overall	0.947	0.943	0.887	0.800
w/o GCF [53]	Group 1	0.924	0.687	0.815	0.750
	Group 2	0.870	0.672	0.891	0.719
	Group 3	0.928	0.909	0.877	0.839
	Overall	0.919	0.905	0.866	0.784
w/o CL	Group 1	0.928	0.692	0.815	0.755
	Group 2	0.872	0.677	0.890	0.722
	Group 3	0.930	0.914	0.876	0.845
	Overall	0.921	0.915	0.866	0.786
Ours	Group 1	0.951	0.720	0.838	0.783
	Group 2	0.895	0.705	0.914	0.750
	Group 3	0.958	0.946	0.899	0.873
	Overall	0.957	0.943	0.889	0.815

In addition, we conducted experiments in a single-source setting, where GFE-CL was trained and tested individually on Groups 1 and 2 of the MuFake dataset, as well as on the TALLIP dataset. As shown in Fig. 4, GFE-CL outperformed other models across all test groups. Particularly on the TALLIP dataset, GFE-CL achieved significantly higher accuracy and F1 scores compared to the closest competitor, demonstrating its outstanding performance. These results further validate the high efficiency of the GFE-CL model in single-source fake news detection and highlight its strong adaptability across different data environments.

5.3. Ablation study

Effectiveness of Each Component. To thoroughly evaluate the roles of key components within the GFE-CL model, we designed the following ablation experiments: (1) GFE-CL w/o DTW for local features—completely removing the Dynamic Time Warping (DTW) [52] module, but replaced by a projection layer; (2) GFE-CL with CTC for Local Features—replacing DTW with Connectionist Temporal Classification (CTC) [37]; (3) GFE-CL w/o Global Context Features (GCF) [53]—eliminating the global context features module, including the cross-source sample repository and attention-based feature enhancement; (4) GFE-CL w/o Dual-level Contrastive Learning (CL)—removing the local-global contrastive learning module from the model (see Table 3).

The experimental results indicate that the removal or replacement of each component significantly impacted the model's performance.

Table 4

Ablation study on the effectiveness of loss function.

Methods	Test	MCFEND		MuFake	
		Acc	F1	Acc	F1
w/o $\mathcal{L}_D$	Group 1	0.936	0.709	0.826	0.772
	Group 2	0.884	0.693	0.903	0.736
	Group 3	0.940	0.931	0.884	0.858
	Overall	0.939	0.934	0.878	0.804
	Unseen(all)	0.874	0.853	0.852	0.787
w/o $\mathcal{L}_{local}$	Group 1	0.933	0.701	0.820	0.765
	Group 2	0.876	0.687	0.896	0.732
	Group 3	0.935	0.924	0.881	0.855
	Overall	0.934	0.928	0.878	0.805
	Unseen(all)	0.837	0.851	0.818	0.786
w/o $\mathcal{L}_{global}$	Group 1	0.931	0.707	0.821	0.769
	Group 2	0.879	0.690	0.896	0.735
	Group 3	0.932	0.926	0.881	0.862
	Overall	0.933	0.929	0.871	0.801
	Unseen(all)	0.861	0.843	0.839	0.778
w/o $\mathcal{L}_{local} \& \mathcal{L}_{global}$	Group 1	0.926	0.706	0.816	0.770
	Group 2	0.871	0.691	0.894	0.738
	Group 3	0.931	0.921	0.875	0.862
	Overall	0.929	0.926	0.868	0.804
	Unseen(all)	0.855	0.837	0.833	0.769
Full loss	Group 1	0.951	0.720	0.838	0.783
	Group 2	0.895	0.705	0.914	0.750
	Group 3	0.958	0.946	0.899	0.873
	Overall	0.957	0.943	0.889	0.815
	Unseen(all)	0.898	0.867	0.868	0.802

GFE-CL w/o DTW exhibited a substantial performance decline without any temporal alignment, underscoring the critical role of DTW in synchronizing multimodal data. GFE-CL with CTC [37], although using a different temporal alignment strategy, performed slightly inferior to DTW [52]. This could be attributed to CTC's alignment assumption that all input sequences are directly mapped to outputs, overlooking potential sequence discrepancies among inputs, which may lead to suboptimal alignment in multimodal contexts. Additionally, the performance decrease in GFE-CL w/o Global Context Features highlighted the importance of these enhancement mechanisms in improving the model's deep semantic understanding of news content. The final results of the GFE-CL w/o Contrastive Learning underscore the importance of contrastive learning in enhancing fake news detection. By optimizing the distance between positive and negative samples, this method significantly bolsters the model's ability to discern real from fake news, showcasing its generalizability across variant sources.

Effectiveness of different loss functions. To evaluate the roles of different loss functions, we designed a series of experiments that involved disabling the domain adaptation loss ($\mathcal{L}_D$), local contrastive loss ($\mathcal{L}_{local}$), and global contrastive loss ($\mathcal{L}_{global}$) within the model. The experimental results, as shown in the Table 4, indicate that the removal

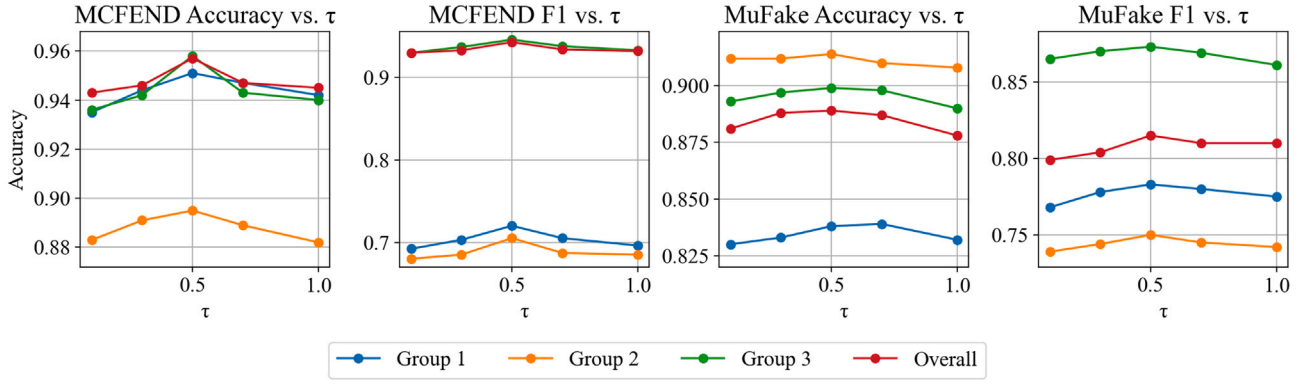


Fig. 5. Performance comparison across various group settings for different τ values on the MCFEND and MuFake datasets.

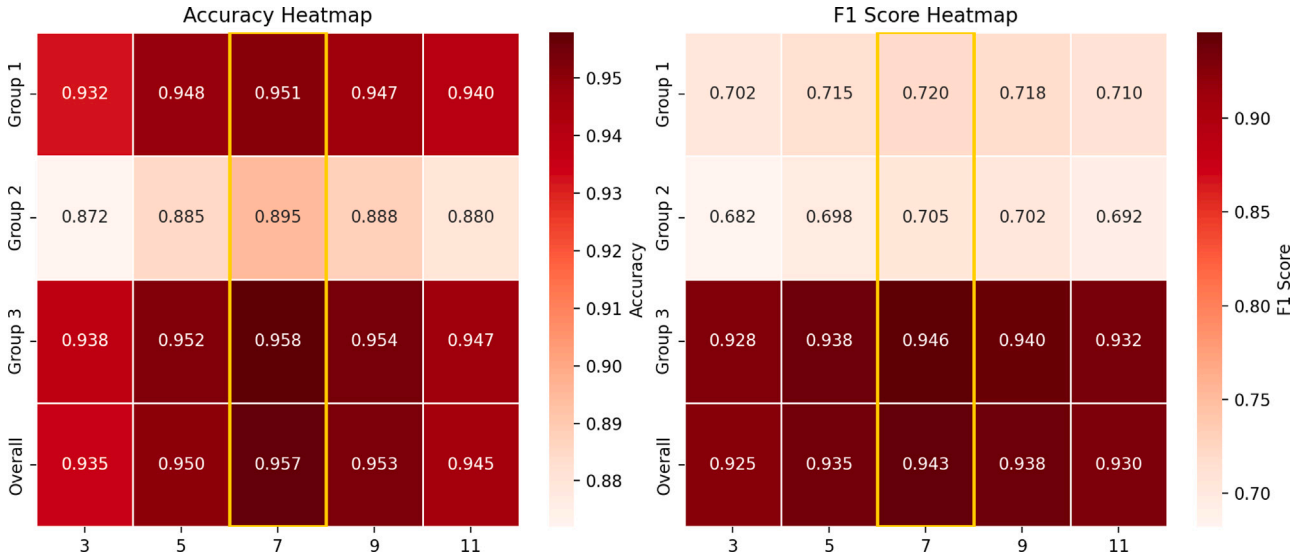


Fig. 6. Performance heatmap of various Top-k values on the MCFEND dataset, with the optimal setting ($k = 7$) highlighted by a golden border. Darker colors indicate lower performance, while brighter colors represent better results.

of each type of loss significantly impacted the overall performance of the model. After removing the cross-source adaptation loss, the model's performance dropped across different and unseen data sources, with an average decline of 2.35% in the overall Acc score, underscoring the importance of the domain adaptation loss in helping the model overcome variations between different sources. Disabling the local contrastive loss led to the most significant performance decline on single-source data, with an overall F1 score decrease of 1.1%, highlighting the crucial role of local contrastive learning in enhancing the model's ability to capture details. The removal of the global contrastive loss resulted in a noticeable decline in the model's ability to handle consistency across data sources and unseen sources, with an overall Acc score decrease of 2.7%, emphasizing the central role of global contrastive learning in maintaining consistency across sources and deepening the understanding of semantic structures.

The Impact of Temperature Parameter τ . To validate the choice of the temperature parameter $\tau = 0.5$, we compared the impact of different τ values ($\tau \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$) on model performance using the MCFEND and MuFake. Fig. 5 presents the performance comparison across different group settings, showing how accuracy and F1 scores vary with τ .

The experimental results indicate that $\tau = 0.5$ consistently yields the best or nearly best performance across all groups, particularly on the MCFEND dataset, where the overall accuracy is approximately 1.1% and 1.0% higher than for $\tau = 0.1$ and $\tau = 1.0$, respectively. Moreover,

the performance variation for τ values between 0.3 and 0.7 is minimal (with fluctuations around $\pm 0.8\%$), indicating that $\tau = 0.5$ lies within a stable advantage range. Theoretically, smaller τ values (such as 0.1) may excessively penalize hard negative samples, while larger values (such as 1.0) may weaken the discriminative power of contrastive learning. Thus, $\tau = 0.5$ effectively balances the gradient contributions from positive and negative samples, making it an ideal choice.

The Impact of Top-k Selection. To validate the choice of Top k , we compared the performance of different Top-k values ($k \in \{3, 5, 7, 9, 11\}$) on the MCFEND dataset, focusing on the differences in accuracy and F1 score across groups (1–3).

The heatmap (Fig. 6) visually presents the performance comparison for different Top-k values. The results show that $k = 7$ (highlighted with a golden border) maintains optimal performance across most groups, with particularly strong performance in Group 3 and Overall. Performance significantly decreases when k is too small (e.g., $k = 3$) or too large (e.g., $k = 11$). This indicates that $k = 7$ effectively balances sample diversity and noise control, making it the best choice for cross-source sample set construction.

5.4. Visualization of fake news detection

We present the case studies in Fig. 7. Specifically, the model performs excellently in technical (e.g., “Audi 3D printing”) and factual contradiction cases (e.g., “Guo Meimei’s low-income application”), effectively capturing the semantic associations between text and image



Fig. 7. Case studies for analyzing the model's performance in fake news detection, including correctly judged factual news, fake news, and misjudged cases.

while ignoring background noise. However, in sensitive topics (such as the misjudgment of Guiyang as the capital of Guangxi) and organizational reporting (Red Cross issuing fake invoices), the model makes errors due to overemphasis on entity co-occurrence and adversarial training biases. Our experiments reveal two main directions for improvement: (1) Enhance contextual modeling of sensitive topics by separating factual and emotional features through adversarial training; (2) Introduce a controversy detector to trigger multi-source verification, improving the robustness of identifying error-correction behaviors and real-name reports.

6. Limitations

Despite its effectiveness, the proposed framework has several limitations that warrant further investigation: (1) Modality Dependence: The model relies on both Dynamic Time Warping (DTW) and contrastive learning for effective alignment across multiple modalities. In scenarios where certain modalities are absent (e.g., text-only news articles), the model's performance may degrade. Future research should explore adaptable architectures capable of handling incomplete or unimodal inputs more robustly. (2) Computational Complexity: The incorporation of global context modeling and dual-level contrastive learning introduces substantial computational overhead, particularly when scaling to large datasets. To enhance scalability and training efficiency, future work should consider the development of lightweight retrieval mechanisms, such as approximate nearest neighbor search or hash-based indexing techniques. (3) Linguistic and Cultural Bias: The datasets used in this study are primarily centered on Chinese and English news content. This focus may limit the model's generalizability to low-resource languages or culturally nuanced information, including regional dialects or idiomatic expressions. Future efforts should evaluate and adapt the model across a broader range of languages and cultural contexts to ensure its cross-linguistic robustness.

7. Conclusion

This study introduces a unified cross-source context enhancement framework aimed at advancing fake news detection across heterogeneous information sources. The proposed approach integrates a cross-source global context learning module and a dual-level contrastive learning strategy to effectively exploit both intra-source and inter-source contextual cues. By leveraging a context-augmented transformer for long-range dependency modeling and applying contrastive learning to align feature representations across modalities and sources, the framework addresses the key limitations of conventional single-source detection methods. Comprehensive evaluations on publicly available multi-source datasets confirm the superiority of the proposed model, which consistently outperforms existing state-of-the-art techniques and achieves substantial improvements in classification accuracy. These results underscore the framework's capability to generalize across diverse data distributions and media types. Future work will focus on enhancing the model's scalability and adaptability by evaluating its performance on larger-scale datasets and incorporating a broader range of modalities. Such efforts are expected to further strengthen the model's robustness in handling the complexity and variability inherent in real-world, multi-platform misinformation scenarios.

CRedit authorship contribution statement

Ruiting Dai: Writing – review & editing, Supervision, Investigation, Conceptualization. **Haoran Meng:** Writing – review & editing, Investigation, Data curation, Conceptualization. **Zhengdao Yuan:** Writing – review & editing, Methodology, Investigation, Data curation. **Lisi Mo:** Project administration, Methodology, Data curation. **Wenwei Zhu:** Visualization, Software, Resources, Funding acquisition, Formal analysis. **Tao He:** Writing – review & editing, Validation, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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